



Day-20 JSON and API

📌 1. Introduction to JSON

◆ What is JSON?

JSON (JavaScript Object Notation) is a lightweight format used to **store and exchange data**.

👉 Think of it like:

| A digital file where data is stored in **key-value pairs**

◆ Example of JSON

```
{  
  "name": "Mahak",  
  "age": 20,  
  "is_student": true  
}
```

◆ Key Features of JSON

✅ 1. Key-Value Structure

- Data is stored as:

```
"key": "value"
```

✓ 2. Supported Data Types

- String → `"name"`
- Number → `20`
- Boolean → `true/false`
- Array → `[1, 2, 3]`

✓ 3. Nested Objects

- JSON can contain another JSON inside it

```
{  
  "student": {  
    "name": "Mahak",  
    "age": 20  
  }  
}
```

2. JSON vs Python Dictionary

Feature	JSON	Python Dictionary
Quotes	Double quotes only <code>" "</code>	Single or double both
Keys	Must be strings	Can be any data type
Usage	Data exchange	Programming

👉 Important:

- JSON is **data format**
- Dictionary is **Python object**

3. JSON in Python

Python provides a built-in module:

```
import json
```

◆ Convert Python → JSON

```
import json

data= {"name": "Mahak", "age": 20}

json_data=json.dumps(data)
print(json_data)
```

👉 `json.dumps()` → converts dictionary → JSON string

◆ Convert JSON → Python

```
import json

json_data='{"name": "Mahak", "age": 20}'

data=json.loads(json_data)
print(data)
```

👉 `json.loads()` → converts JSON → dictionary

📌 4. Working with JSON Files

◆ Write JSON to File

```
import json

data= {"name": "Mahak", "age": 20}

with open("data.json", "w") as f:
    json.dump(data, f)
```

◆ Read JSON from File

```
import json

with open("data.json", "r") as f:
    data = json.load(f)

print(data)
```

5. What is an API?

◆ Definition

API (Application Programming Interface) allows different applications to **communicate with each other**.

◆ Simple Understanding

👉 Think of API like a **waiter in a restaurant**:

- You → Request (order food)
- Waiter → API
- Kitchen → Server
- Food → Response

6. API Working Process

1. Client sends request
2. Server processes request
3. Server sends response
4. Data usually comes in **JSON format**

7. Making API Calls in Python

We use the **requests library**:

```
pip install requests
```

◆ Example API Call

```
import requests

url="https://api.example.com/data"

response=requests.get(url)

data=response.json()

print(data)
```

◆ Key Points

- `requests.get()` → sends request
- `response.json()` → converts response into Python data

8. JSON and API Connection

👉 Important Concept:

- API returns data in **JSON format**
- Python converts JSON → dictionary
- Then we use it in our program

9. Project: Weather App

◆ What it does:

- Fetch live weather data using API
- Show:
 - Temperature
 - Humidity

- Weather condition
-

♦ **Learning Outcomes:**

- API calling
 - JSON parsing
 - Data extraction
-

10. Project: Currency Converter

♦ **What it does:**

- Converts one currency to another
 - Uses real-time API data
-

♦ **Example Use:**

- USD → INR
 - INR → EUR
-

♦ **Benefits:**

- Real-time data
 - Useful for finance apps
 - Easy to implement
-

11. Advantages of JSON & API

 **Fast Data Exchange**

 **Lightweight**

 **Easy to Understand**

 **Used in Real-World Applications**

 **Works with Almost All Languages**

12. Final Conclusion

- JSON is the **language of data exchange**
 - APIs are the **bridge between applications**
 - Python makes both **simple and powerful**
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13. Key Takeaway for Students

If you understand:

- JSON structure
- API calls
- Data extraction

 You are entering the world of:

- Web development
- Data science
- Real-world applications 